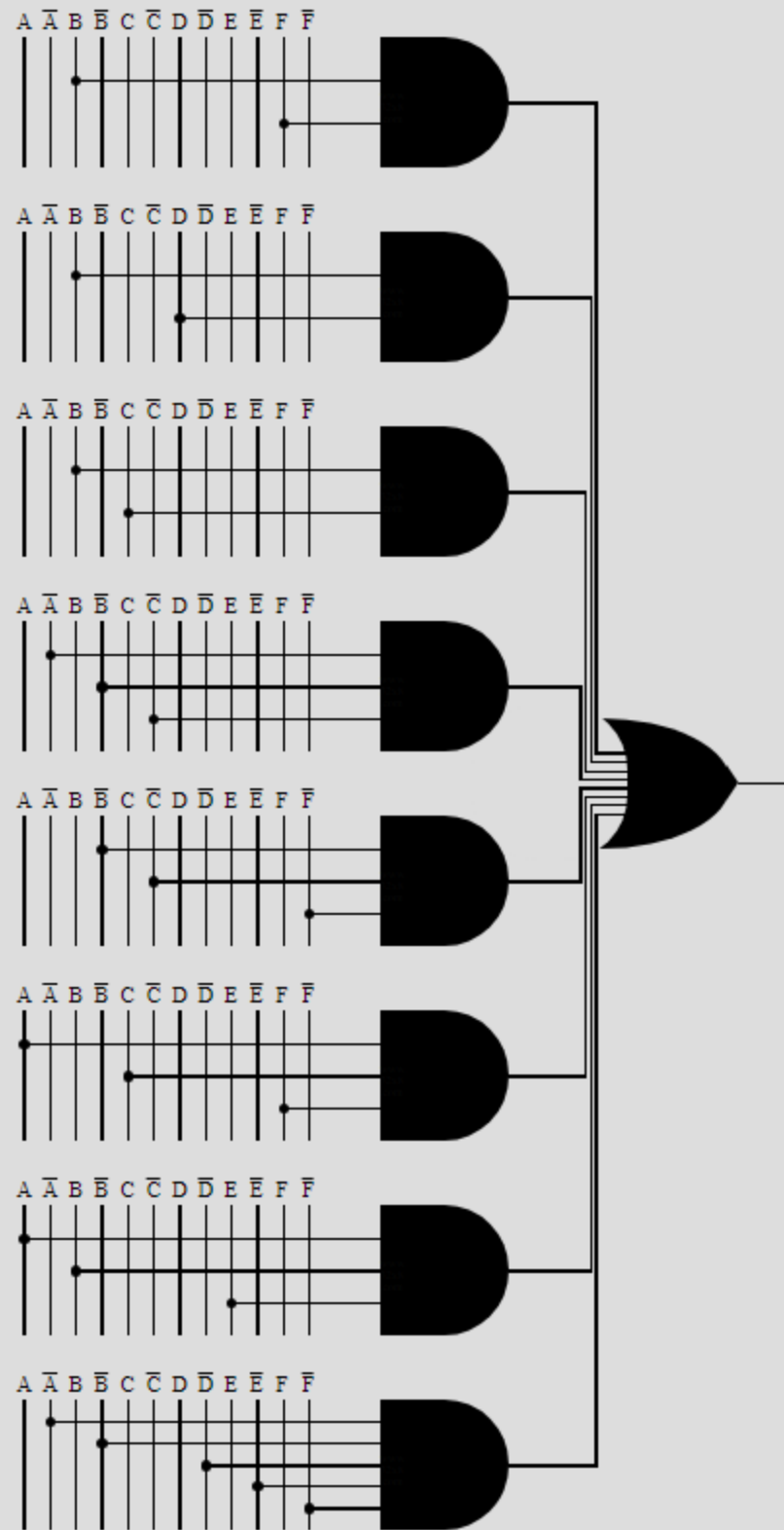


Simplificaciones y Diagrama logico

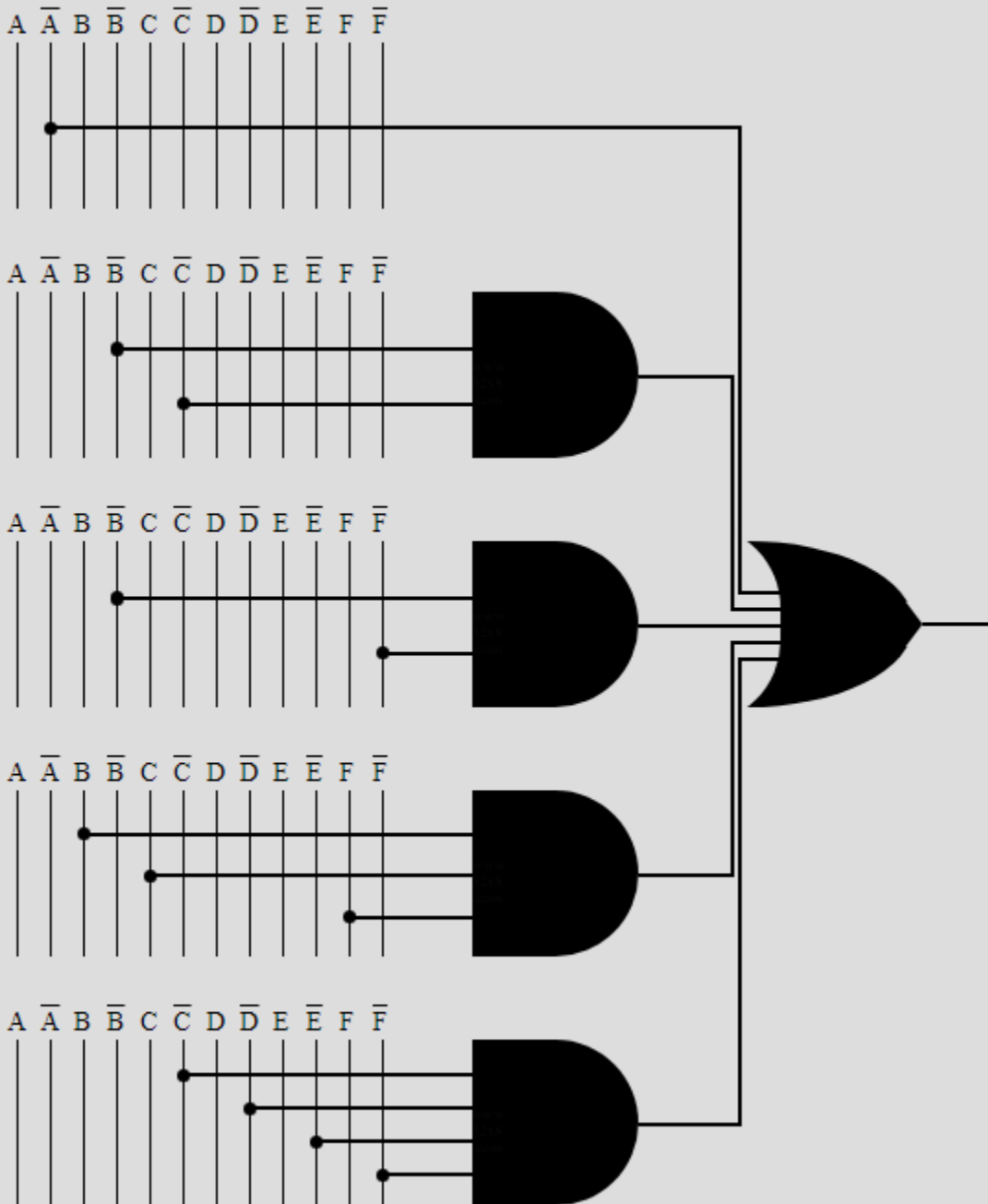
Display 1

$$y = BF + BD + BC + A\bar{B}C + B'CF + ACF + ABE + AB'DEF$$

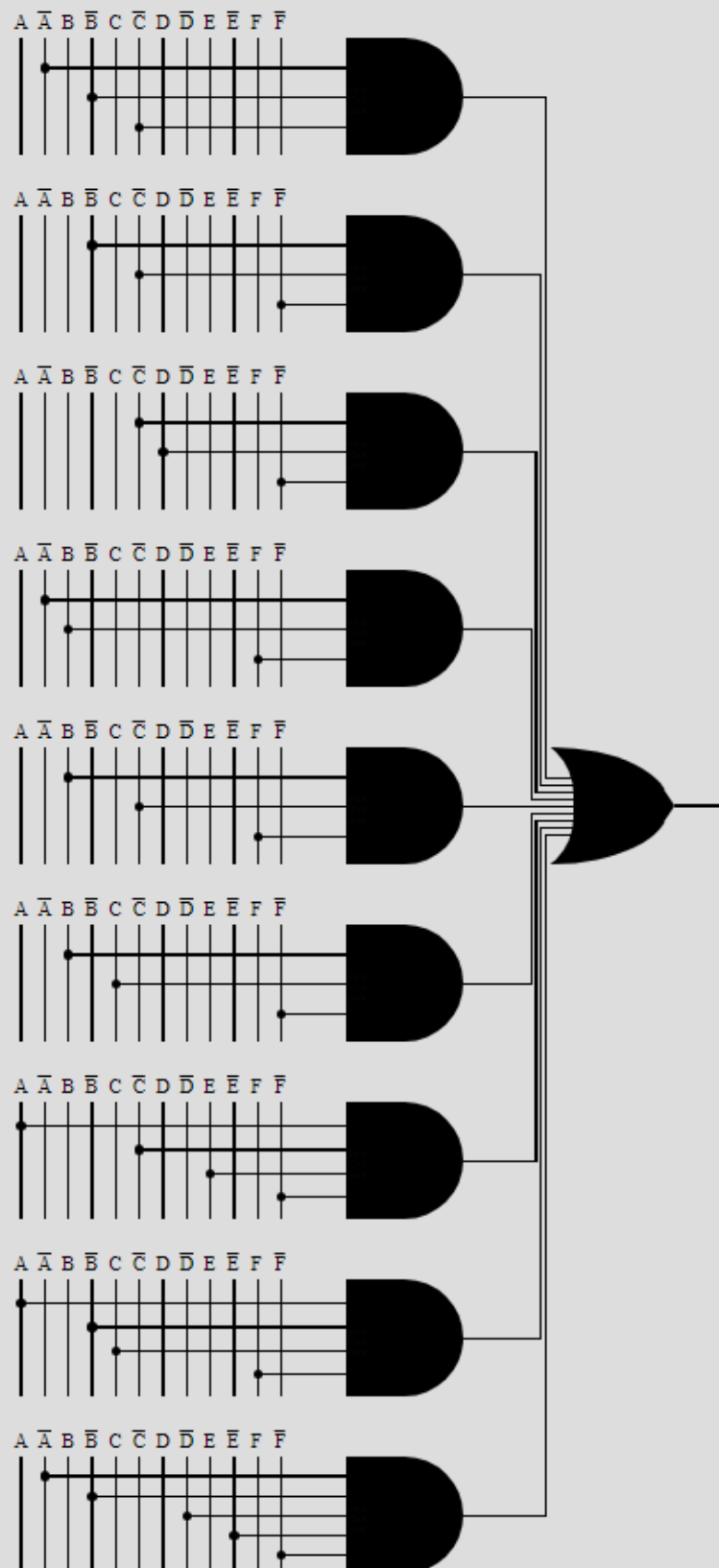


DA1

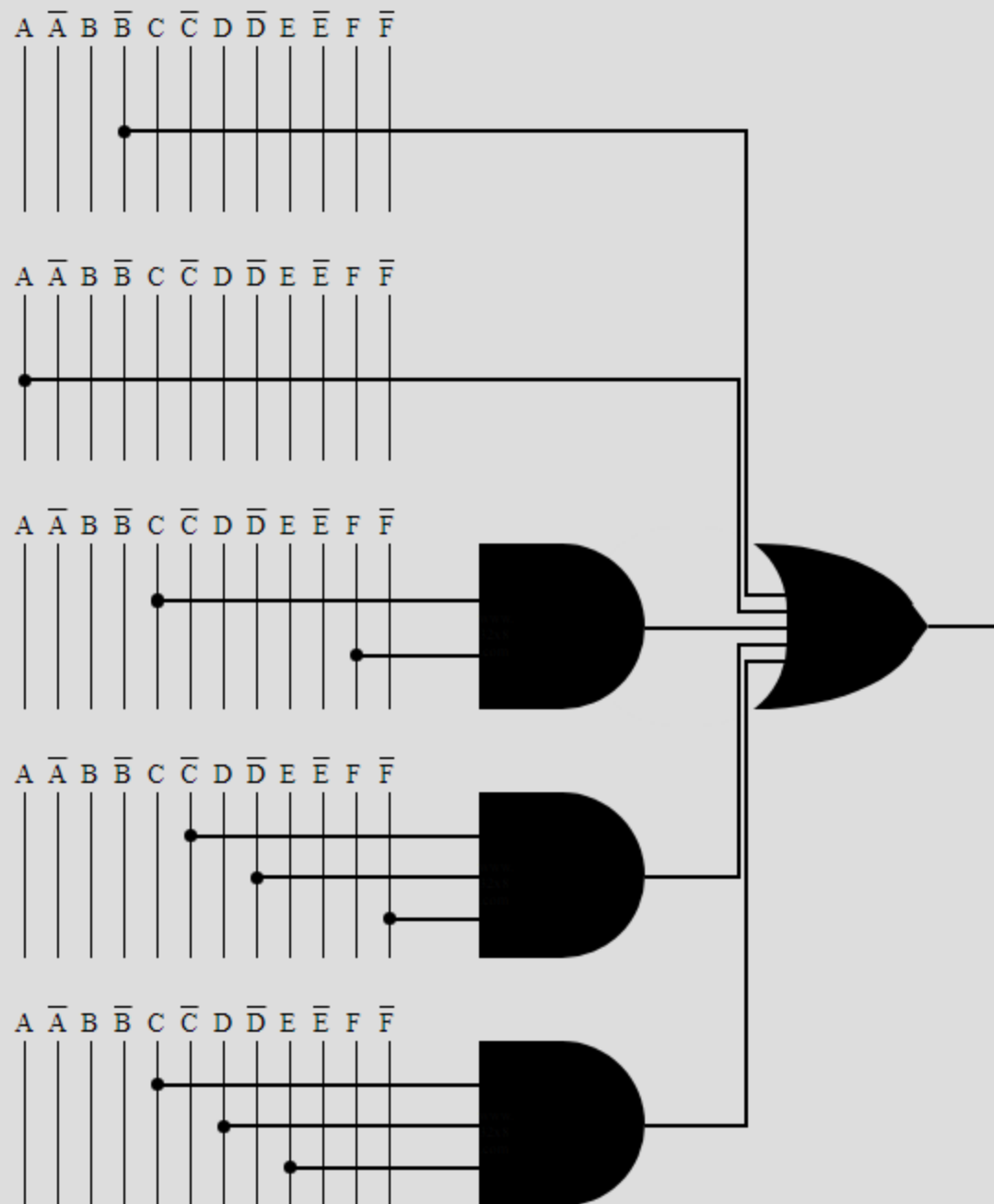
$$y = A' + B'C' + B'F' + BCF + C'D'EF'$$



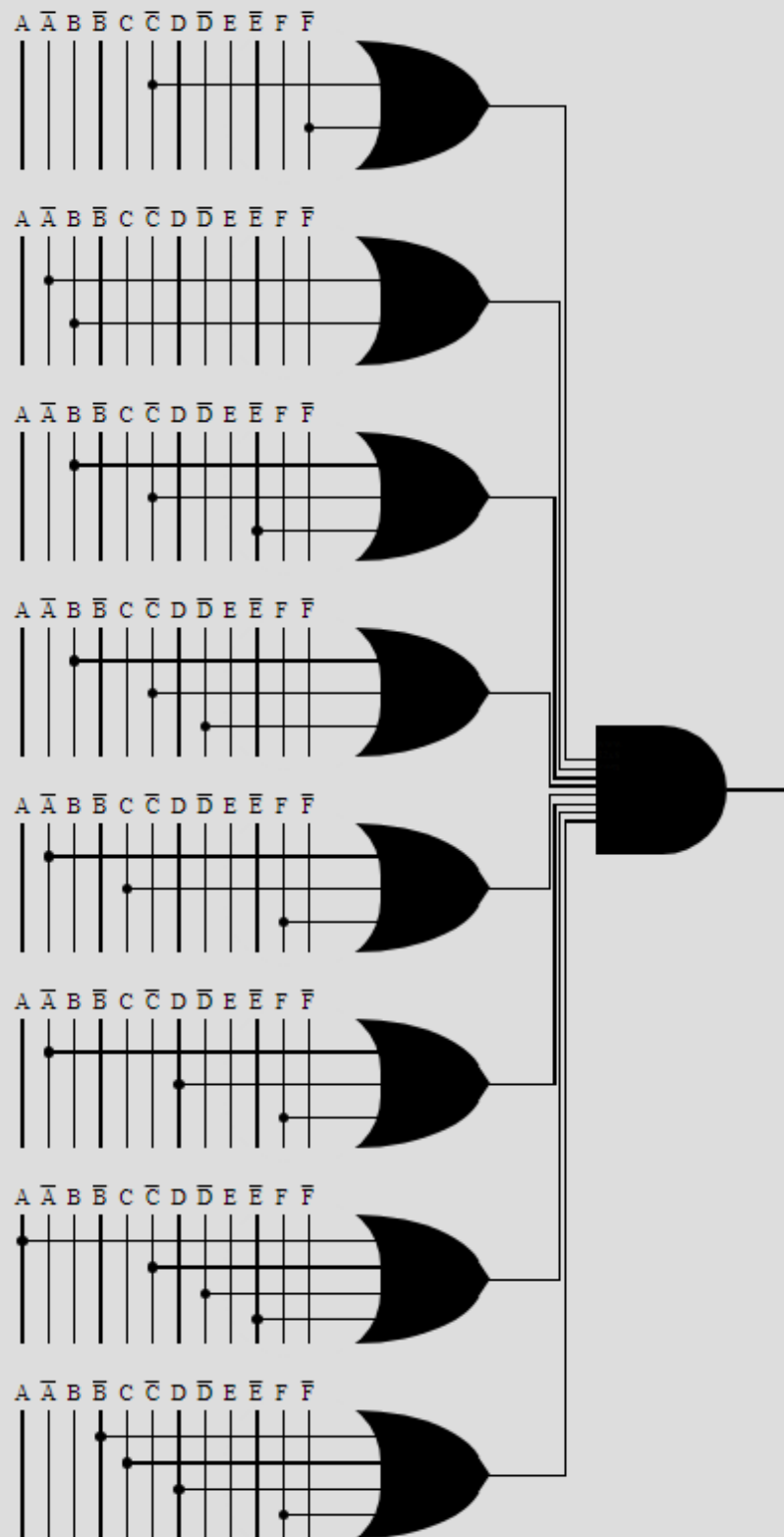
$$y = A'B'C' + B'CF + C'DF + ABF + BCF + BCF + AC'EF + AB'CF + AB'DEF$$



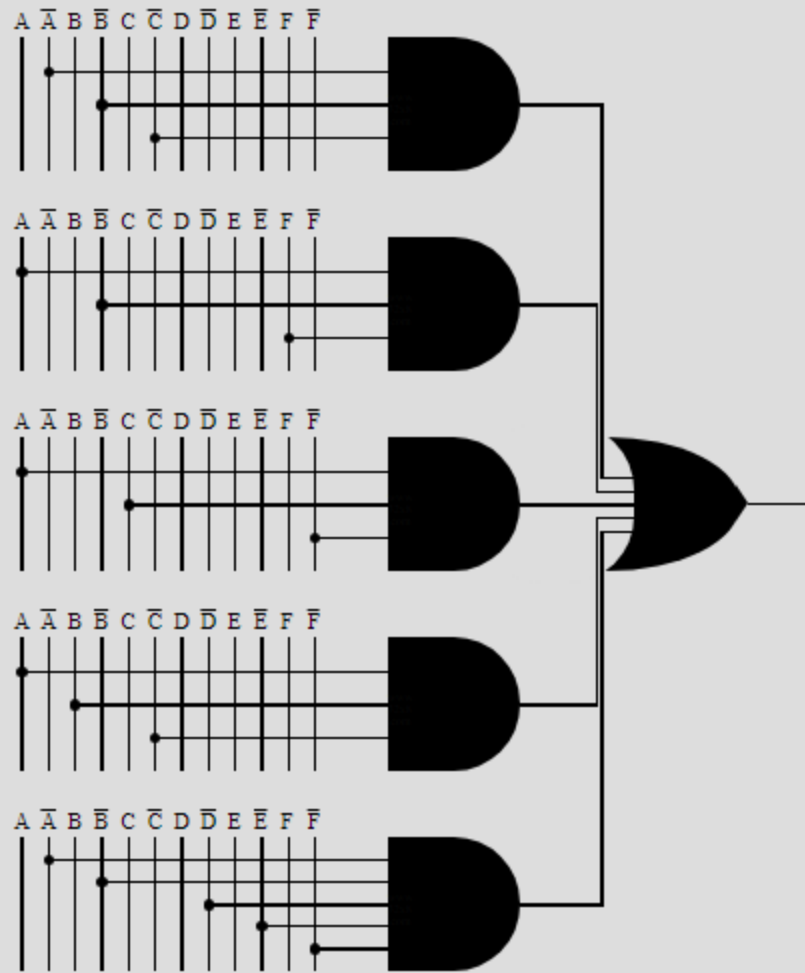
$$y = B' + A + CF + C'DF' + CDE$$



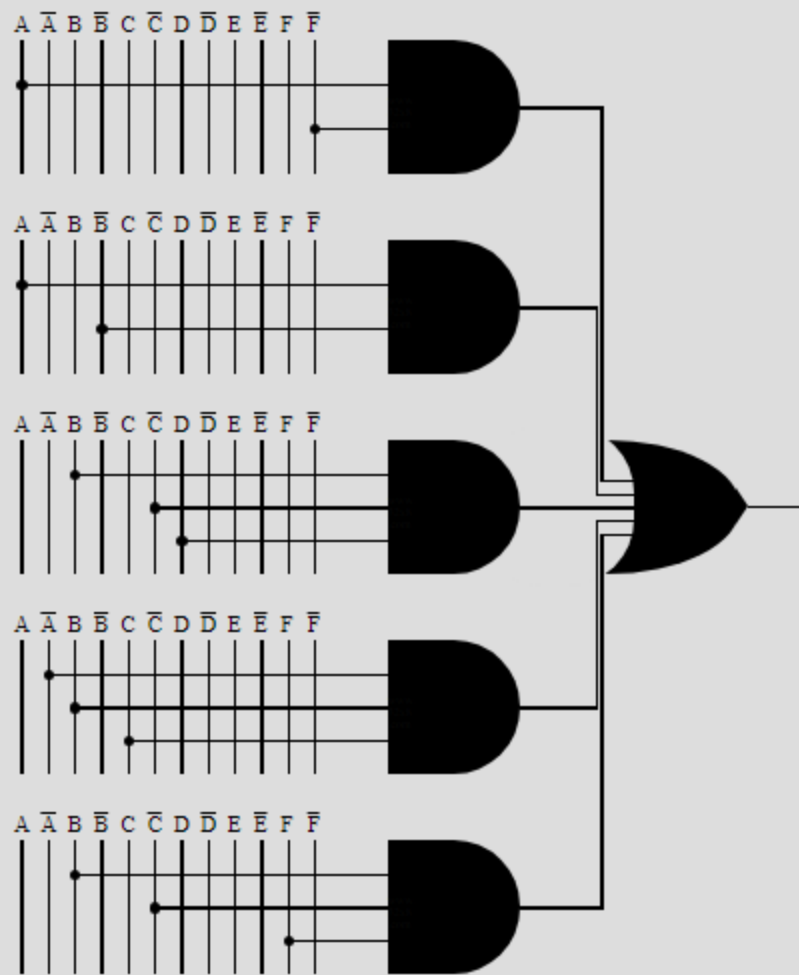
$$y = (C' + F')(A' + B)(B + C' + E)(B + C' + D')(A' + C + F)(A' + D + F)(A + C' + D' + E')(B' + C + D + F)$$



$$y = A'B'C + ABF + ACF + ABC' + AB'D'EF$$

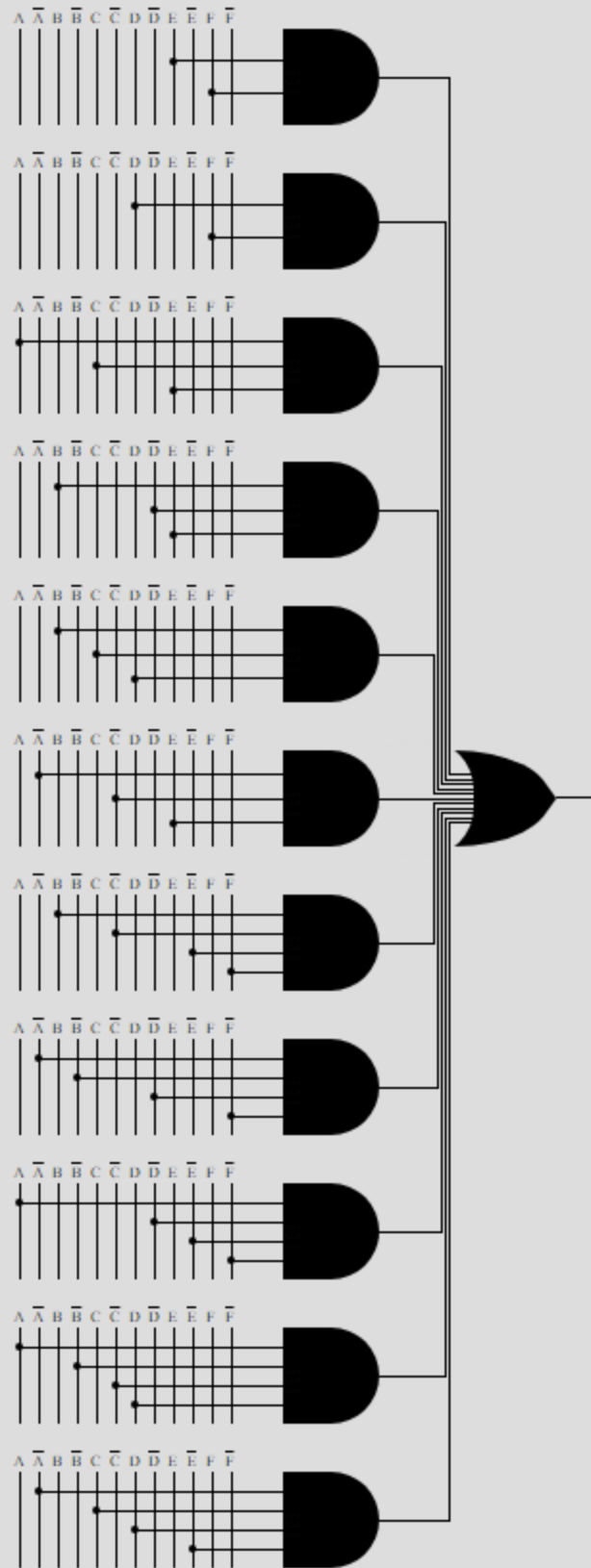


$$y = AF' + AB' + BCD + A'BC + BCF$$

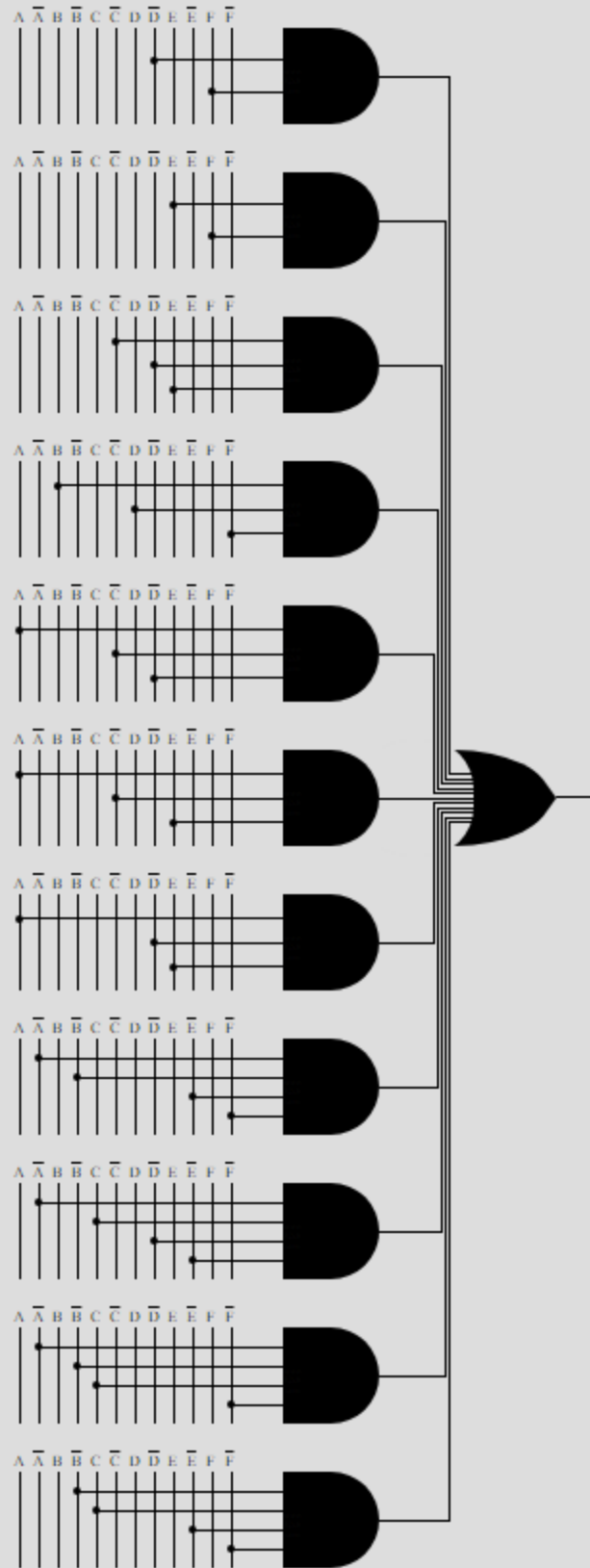


Display 2

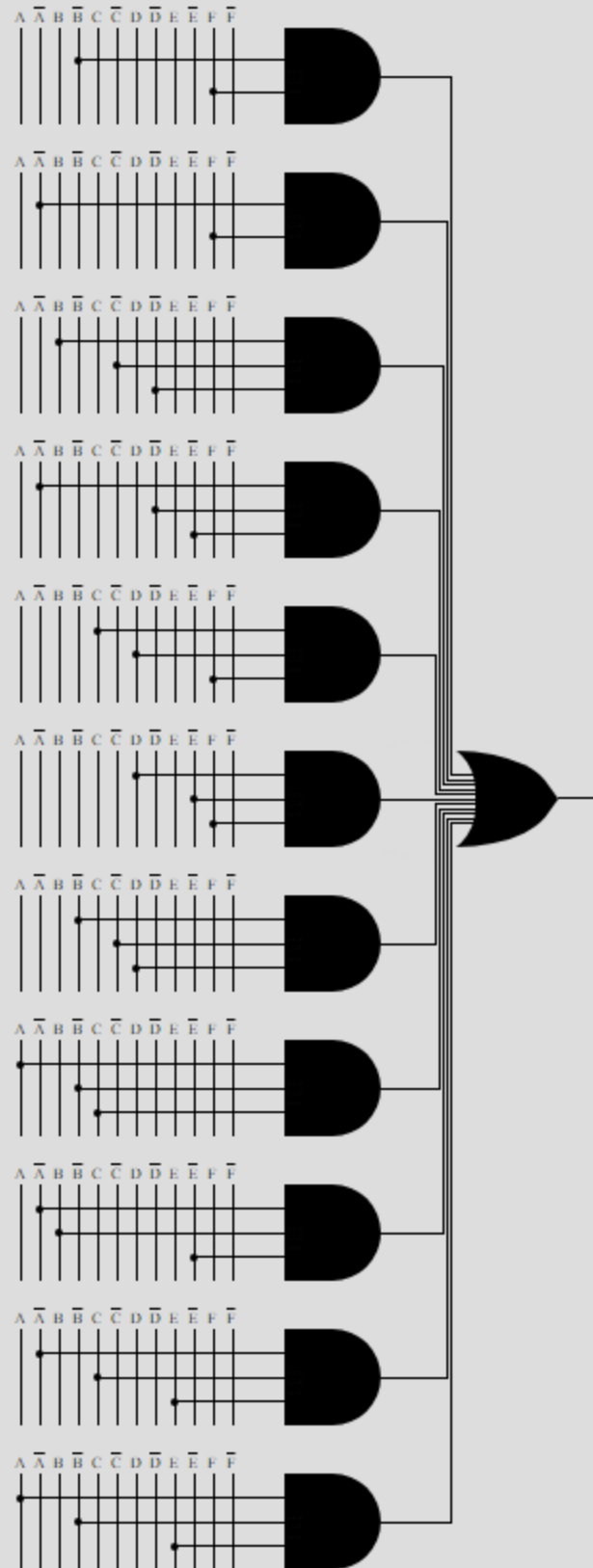
$$y = EF + DF + ACE + BD'E + BCD + A'CE + BC'EF' + A'BD'F' + ADE'F' + AB'CD + A'CDE'$$



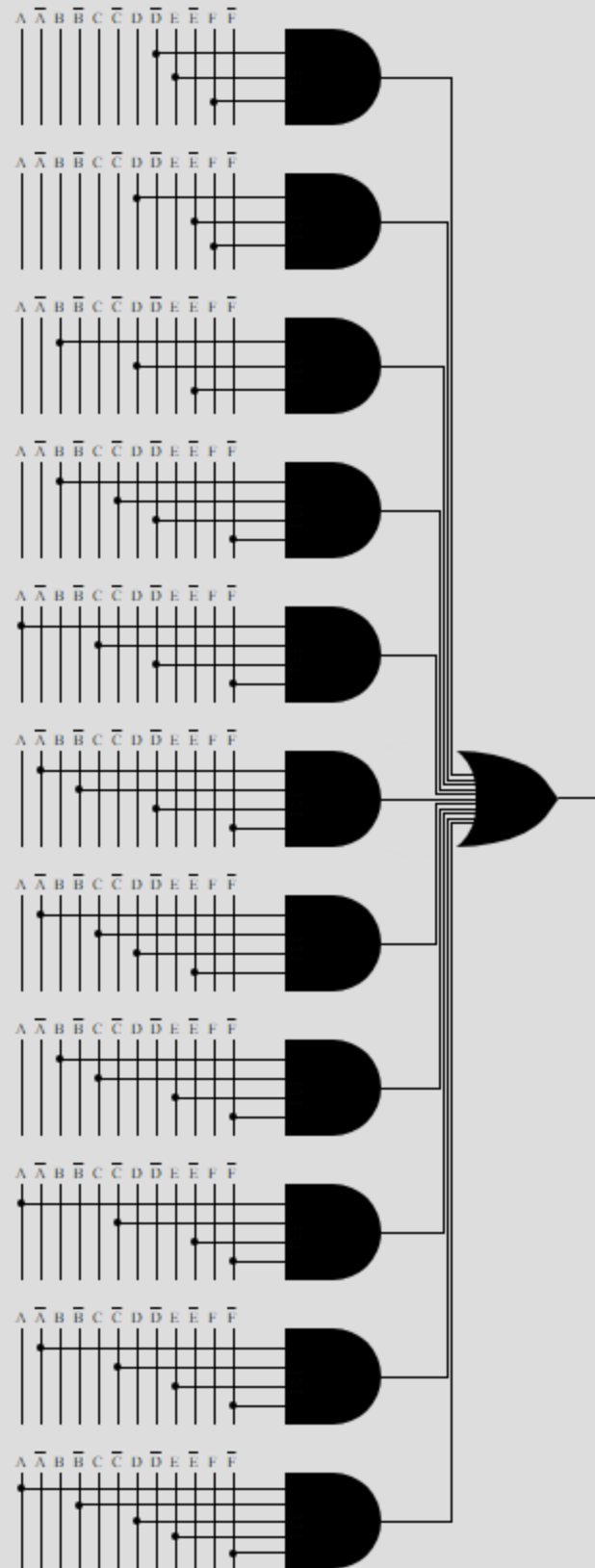
$$y = DF + EF + C'DE + BDF + AC'D + ACE + ADE + A'B'EF + A'CD'E + A'BC'E + B'CE'$$



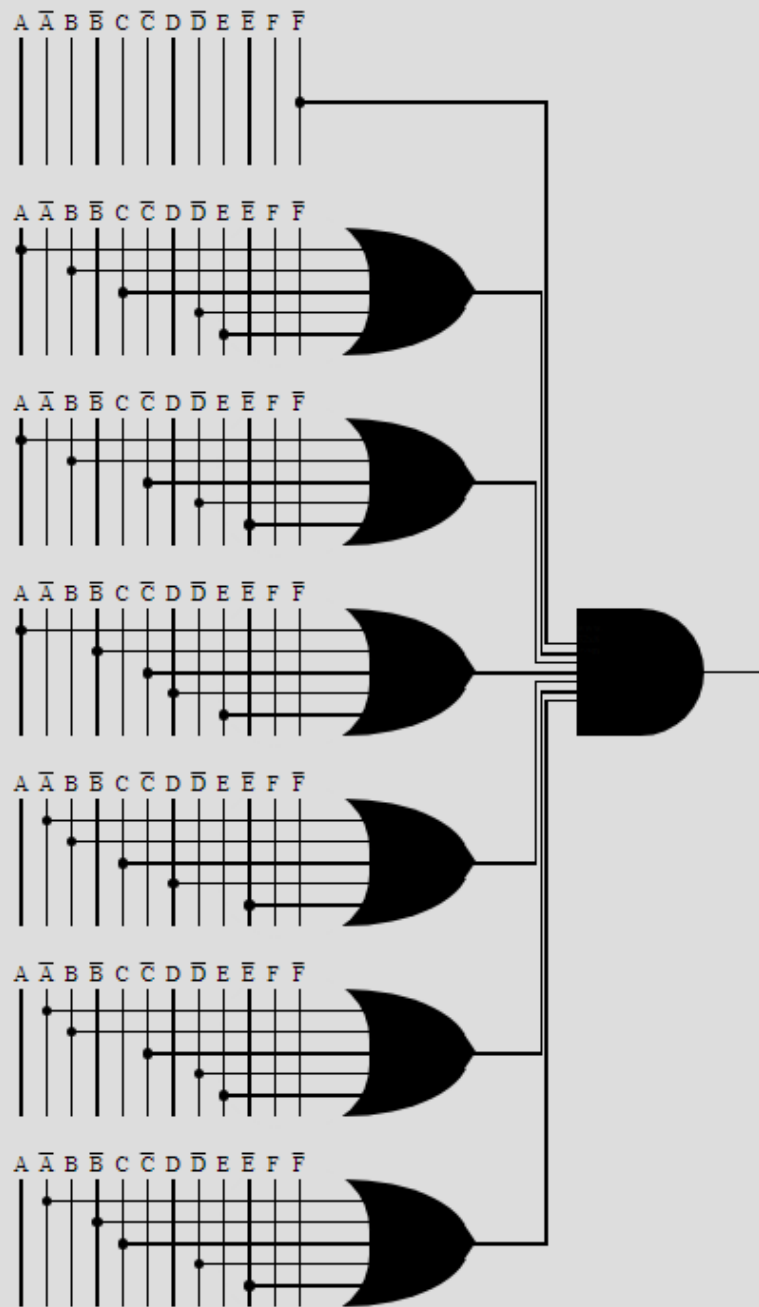
$$y = BF + AF + BCD' + ADE' + CDF + DEF + BCD + ABC + ABE' + ACE + AB'E$$



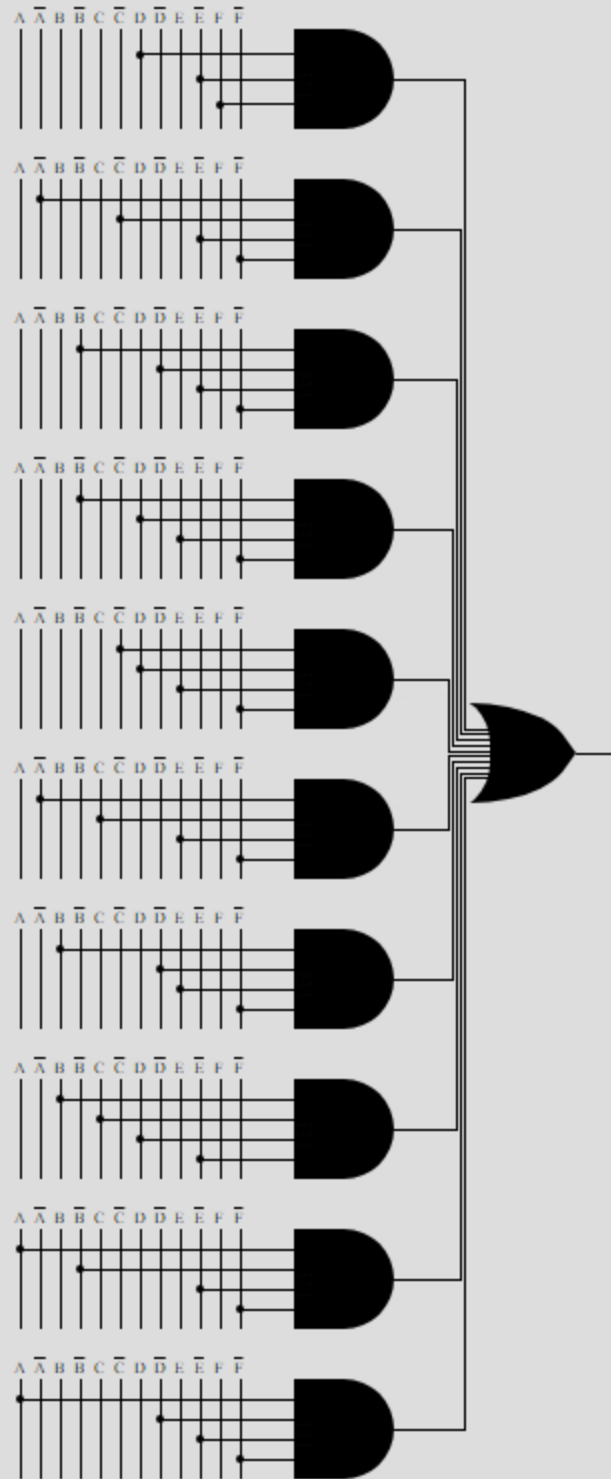
$$y = D'E'F + DE'F + BDE' + BC'D'F + ACDF' + A'B'D'F' + A'CDE' + BCEF' + AC'E'F' + A'C'E'F' + AB'DEF'$$



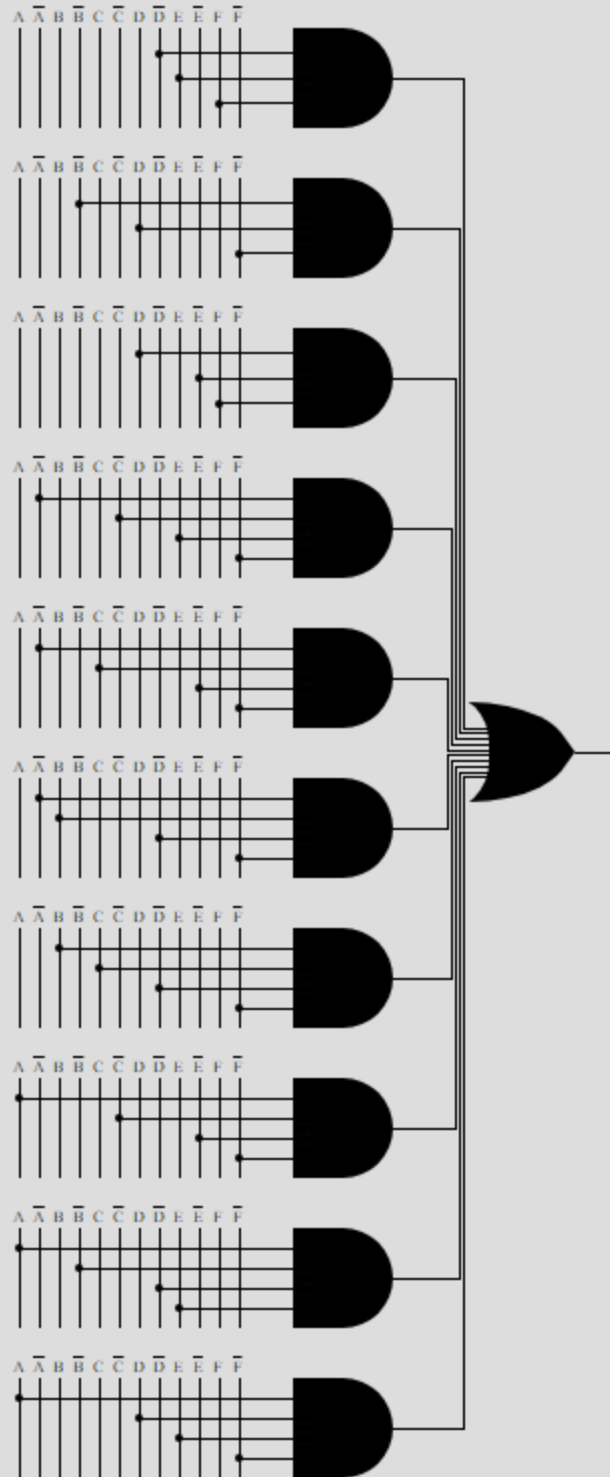
$$y = (\bar{F})(A+B+C+D+E)(A+B+C'+D'+E)(A+B'+C+D+E)(A'+B+C+D+E)(A'+B+C'+D'+E)(A'+B'+C+D'+E)$$



$$y = DEF + A'CE'F + B'D'E'F + B'DE'F + C'DE'F + A'CE'F + B'D'E'F + BCDE' + AB'E'F + AD'E'F$$



$$y = D'EF + B'DF + DE'F + A'CE'F + A'CE'F' + A'BDF' + BCD'F + AC'EF' + AB'DE + ADEF$$



DataSheet y Cotizacion

4081B input: utilice 18

Fully static operation

Wide supply voltage range from 3.0 V to 15.0 V

CMOS low power dissipation

High noise immunity

5 V, 10 V, and 15 V parametric ratings

Standardized symmetrical output characteristics

Complies with JEDEC standard JESD 13-B

ESD protection:

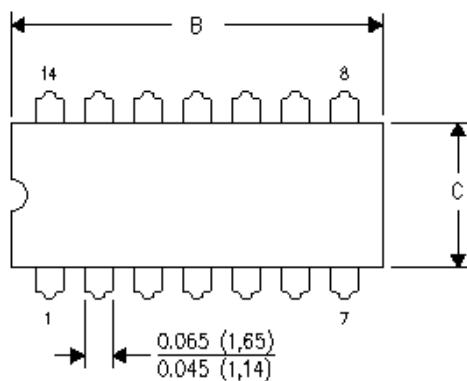
- HBM JESD22-A114F exceeds 2000 V
- MM JESD22-A115-B exceeds 200 V

Specified from -40 °C to +85 °C and -40 °C to +125 °C

Precio por unidad = \$0.120

18 Unidades = \$2.16

4073B input: utilice 26



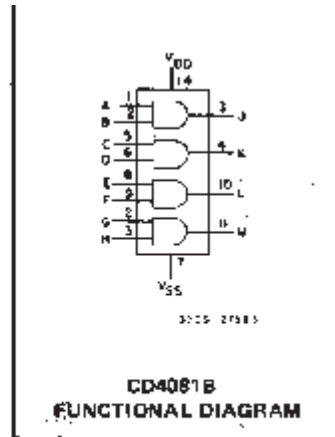
PINS ** DIM	14	16	18	20
A	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC
B MAX	0.785 (19,94)	.840 (21,34)	0.960 (24,38)	1.060 (26,92)
B MIN	—	—	—	—
C MAX	0.300 (7,62)	0.300 (7,62)	0.310 (7,87)	0.300 (7,62)
C MIN	0.245 (6,22)	0.245 (6,22)	0.220 (5,59)	0.245 (6,22)

Precio por unidad: \$0.251

18 precio: \$6.526

4082Binput:utilice 23

- **Medium-Speed Operation** — t_{PLH} ,
 $t_{PHL} = 60 \text{ ns}$ (typ.) at $V_{DD} = 10 \text{ V}$
- **100% tested for quiescent current at 20 V**
- **Maximum input current of $1 \mu\text{A}$ at 18 V over full package-temperature range; 100 nA at 18 V and 25°C**
- **Noise margin (full package-temperature range) =**
 - 1 V at $V_{DD} = 5 \text{ V}$
 - 2 V at $V_{DD} = 10 \text{ V}$
 - 2.5 V at $V_{DD} = 15 \text{ V}$
- **Standardized, symmetrical output characteristics**
- **5-V, 10-V, and 15-V parametric ratings**
- **Meets all requirements of JEDEC Tentative Standard No. 13B, "Standard Specifications for Description of 'B' Series CMOS Devices"**



Precio por unidad= \$0.195

23 unidades = \$4.485

4072B input: utilice 16

CHARACTER- ISTIC	CONDITIONS			LIMITS AT INDICATED TEMPERATURES (°C)							UNITS
	V _O (V)	V _{IN} (V)	V _{DD} (V)	+25							
				-55	-40	+85	+125	Min.	Typ.	Max.	
Quiescent Device Current, I _{DD} Max.	—	0,5	5	0.25	0.25	7.5	7.5	—	0.01	0.25	μA
	—	0,10	10	0.5	0.5	15	15	—	0.01	0.5	
	—	0,15	15	1	1	30	30	—	0.01	1	
	—	0,20	20	5	5	150	150	—	0.02	5	
Output Low (Sink) Current I _{OL} Min.	0.4	0,5	5	0.64	0.61	0.42	0.36	0.51	1	—	mA
	0.5	0,10	10	1.6	1.5	1.1	0.9	1.3	2.6	—	
	1.5	0,15	15	4.2	4	2.8	2.4	3.4	6.8	—	
Output High (Source) Current, I _{OH} Min.	4.6	0,5	5	-0.64	-0.61	-0.42	-0.36	-0.51	-1	—	
	2.5	0,5	5	-2	-1.8	-1.3	-1.15	-1.6	-3.2	—	
	9.5	0,10	10	-1.6	-1.5	-1.1	-0.9	-1.3	-2.6	—	
13.5	0,15	15	-4.2	-4	-2.8	-2.4	-3.4	-6.8	—		
Output Voltage: Low-Level, V _{OL} Max.	—	0,5	5	0.05				—	0	0.05	V
	—	0,10	10	0.05				—	0	0.05	
	—	0,15	15	0.05				—	0	0.05	
Output Voltage: High-Level, V _{OH} Min.	—	0,5	5	4.95				4.95	5	—	
	—	0,10	10	9.95				9.95	10	—	
	—	0,15	15	14.95				14.95	15	—	
Input Low Voltage, V _{IL} Max.	0.5, 4.5	—	5	1.5				—	—	1.5	V
	1, 9	—	10	3				—	—	3	
	1.5, 13.5	—	15	4				—	—	4	
Input High Voltage, V _{IH} Min.	4.5	—	5	3.5				3.5	—	—	
	9	—	10	7				7	—	—	
	13.5	—	15	11				11	—	—	
Input Current I _{IN} Max.		0,18	18	±0.1	±0.1	±1	±1	—	±10 ⁻⁵	±0.1	μA

Precio por unidad= \$0.239

16 unidades= \$3.824

4071B input: utilice 14

Symbol	Parameter	Value	Unit
V _{DD} *	Supply Voltage : HCC Types	- 0.5 to + 20	V
	HCF Types	- 0.5 to + 18	V
V _i	Input Voltage	- 0.5 to V _{DD} +0.5	V
I _i	DC Input Current (any one input)	± 10	mA
P _{tot}	Total Power Dissipation (per package)	200	mW
	Dissipation per Output Transistor for Top = Full Package-temperature Range	100	mW
T _{op}	Operating Temperature : HCC Types	- 55 to + 125	°C
	HCF Types	-40 to + 85	°C
T _{stg}	Storage Temperature	- 65 to + 150	°C

Precio por unidad = \$0.145

14 unidades = \$2.03

4075B input: utilice 24

*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
CD4071BM96	SOIC	D	14	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
CD4071BNSR	SO	NS	14	2000	330.0	16.4	8.2	10.5	2.5	12.0	16.0	Q1
CD4071BPWR	TSSOP	PW	14	2000	330.0	12.4	7.0	5.6	1.6	8.0	12.0	Q1
CD4072BM96	SOIC	D	14	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
CD4072BNSR	SO	NS	14	2000	330.0	16.4	8.2	10.5	2.5	12.0	16.0	Q1
CD4075BM96	SOIC	D	14	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
CD4075BNSR	SO	NS	14	2000	330.0	16.4	8.2	10.5	2.5	12.0	16.0	Q1
CD4075BPWR	TSSOP	PW	14	2000	330.0	12.4	7.0	5.6	1.6	8.0	12.0	Q1

Precio por unidad = \$0.200

24 unidades = \$4.8

Display de 7 segmentos: utilice 2

Colour	Red	Red	Red	Green	Yellow	
Colour Code	R	E [†]	S	G	Y	
Brightness Category	Bright	Extra Bright	Super Bright	Extra Bright	Extra Bright	
Typical Luminous Intensity IF=10mA	700	3500	17500	4000	3500	∞cd
Peak Wavelength IF=20mA	697	630	660	565	585	nm
Typical Forward Voltage IF=20mA	2.1	2.1	2.1	2.2	2.1	Volts
Maximum Forward Voltage IF=20mA	2.8	2.8	2.8	2.8	2.8	Volts
Maximum Reverse Current VR=5V	100	100	100	100	100	∞A

Precio por unidad = \$2.70

2 unidades = \$4.40

El precio total para armar el circuito seria
aproximadamente \$28.35 dolares